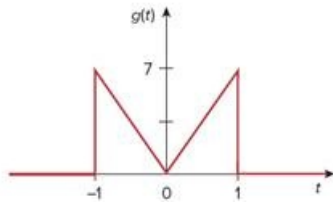


Problem

Find the Fourier transform of the waveform shown in Fig. 18.29.

Figure 18.29



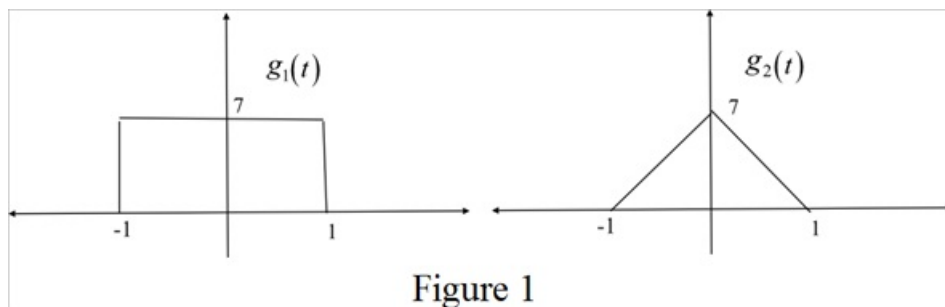
Step-by-step solution

Step 1 of 5

Refer to Figure 18.29, in the text book.

To obtain solution for figure 18.29, write $g(t) = g_1(t) - g_2(t)$ (1)

The signals $g_1(t)$ and $g_2(t)$ are shown in figure 1.



Step 2 of 5

The Fourier transform of $g_1(t)$ is,

$$\begin{aligned}
 G_1(j\omega) &= \int_{-1}^1 7e^{-j\omega t} dt \\
 &= \frac{7}{-j\omega} \left[e^{-j\omega t} \right]_{-1}^1 \\
 &= \frac{7}{-j\omega} (e^{-j\omega} - e^{j\omega}) \\
 &= \frac{7}{j\omega} \left(\frac{e^{j\omega} - e^{-j\omega}}{2j} \right) (2j) \\
 G_1(j\omega) &= 14 \frac{\sin \omega}{\omega} \\
 &= 14 \operatorname{sinc} \omega
 \end{aligned}$$

Step 3 of 5

The Fourier transform of $g_2(t)$ can be found by using derivative property as shown in figure 2.

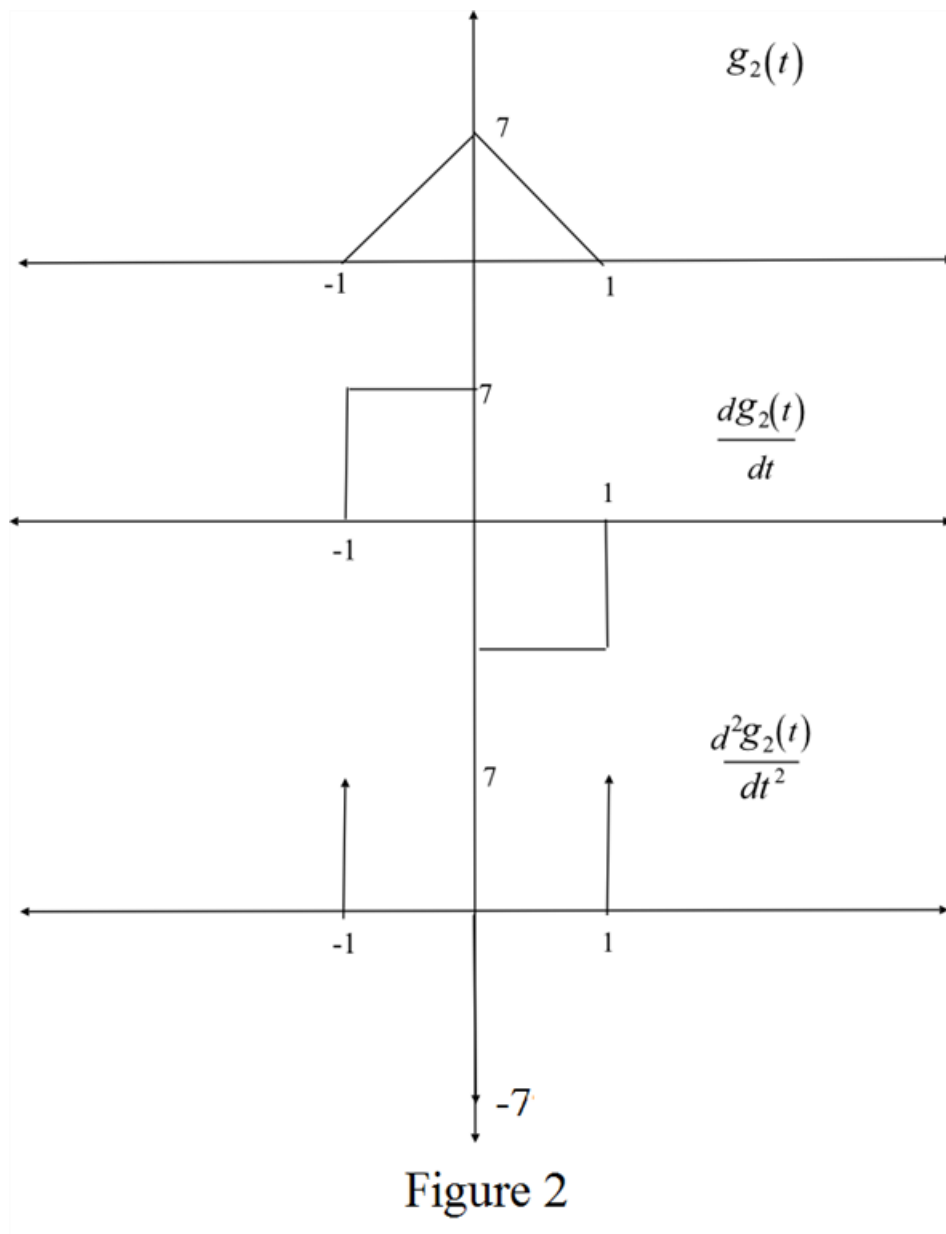


Figure 2

Step 4 of 5

Figure 2 shows the signals obtained by successive differentiation of $g_2(t)$.

The signal $\frac{d^2 g_2(t)}{dt^2}$ can be expressed as,

$$\frac{d^2 g_2(t)}{dt^2} = 7[\delta(t+1) - 2\delta(t) + \delta(t-1)]$$

Using time differentiation property of Fourier transform,

$$F\left[\frac{d^2 g_2(t)}{dt^2}\right] = 7F[\delta(t+1) - 2\delta(t) + \delta(t-1)]$$

$$\begin{aligned} (j\omega)^2 G_2(j\omega) &= 7(e^{j\omega} - 2 + e^{-j\omega}) \quad (\because x(t+1) \leftrightarrow e^{j\omega} X(j\omega)) \\ &= 7[2\cos\omega - 2] \\ &= 14[\cos\omega - 1] \end{aligned}$$

$$\begin{aligned} G_2(j\omega) &= \frac{14[\cos\omega - 1]}{(j\omega)^2} \\ &= \frac{14[1 - \cos\omega]}{\omega^2} \end{aligned}$$

Apply Fourier transform to the equation (1).

$$G(j\omega) = G_1(j\omega) - G_2(j\omega)$$

Substitute $\frac{14[1 - \cos \omega]}{\omega^2}$ for $G_2(j\omega)$ and $14 \sin c \omega$ for $G_1(j\omega)$.

$$G(j\omega) = 14 \sin c \omega - \frac{14[1 - \cos \omega]}{\omega^2}$$

Therefore, the Fourier transform of $g(t)$ is,

$$G(j\omega) = 14 \sin c \omega - \frac{14[1 - \cos \omega]}{\omega^2}.$$